

*MA.6.NSO.2.3***Benchmark**

MA.6.NSO.2.3 Solve multi-step real-world problems involving any of the four operations with positive multi-digit decimals or positive fractions, including mixed numbers.

Benchmark Clarifications:

Clarification 1: Within this benchmark, it is not the expectation to include both decimals and fractions within a single problem.

Connecting Benchmarks/Horizontal Alignment

- MA.6.AR.1.1
- MA.6.AR.2.4
- MA.6.AR.3.2, MA.6.AR.3.3, MA.6.AR.3.4, MA.6.AR.3.5
- MA.6.GR.2.2, MA.6.GR.2.3, MA.6.GR.2.4
- MA.6.DP.1.2, MA.6.DP.1.3, MA.6.DP.1.4, MA.6.DP.1.6
- MA.7.NSO.2.3

Terms from the K-12 Glossary

- Area Model
- Commutative Property of Multiplication
- Dividend
- Divisor
- Expression

Vertical Alignment**Previous Benchmarks**

- MA.5.AR.1.1, MA.5.AR.1.2, MA.5.AR.1.3
- MA.5.M.2.1
- MA.5.GR.2.1

Next Benchmarks

- MA.8.NSO.1.6, MA.8.NSO.1.7

Purpose and Instructional Strategies

In previous courses, students solved multi-step real-world problems involving the four operations with whole numbers as well as addition, subtraction and multiplication for solving real-world problems with fractions and for solving problems with decimals involving money, area and perimeter. In grade 6 accelerated, students solve multi-step real-world problems with positive fractions and decimals. Students will also solve real-world problems involving any of the four operations with positive and negative rational numbers. In future courses, students will use the order of operations with rational numbers, including exponents and radicals to solve multi-step mathematical and real-world problems.

- This benchmark is the culmination of MA.6.NSO.2. It is built on the skills found in MA.6.NSO.2.1 and MA.6.NSO.2.2, so instruction provides practice of these skills within real-world contexts (*MTR.5.1, MTR.7.1*).

- Instruction includes engaging in questions such as:
 - What do you know from the problem?
 - What is the problem asking you to find?
 - Are you putting groups together? Taking groups apart? Or both?
 - Are the groups you are working with the same sizes or different sizes?
 - Can you create a visual model to help you understand or see patterns in your problem?
- Students should have experience using technology with decimals and fractions as they occur in the real world (*MTR.7.1*). This benchmark directly connects to MA.7.NSO.2.3 where students will be solving real-world problems with all rational numbers.
- Students should develop fluency with and without the use of a calculator when performing operations with positive rational numbers.

Common Misconceptions or Errors

- Students may incorrectly oversimplify a problem by mechanically circling the numbers, underlining the question, and boxing in keywords and then jumping to an answer, or procedure, without taking the time to comprehend the context or situation (*MTR.2.1, MTR.4.1, MTR.5.1, MTR.7.1*).
- Students may incorrectly apply rules for adding or subtracting decimals to multiplication of decimals, believing place values must be aligned.
- Students may confuse the lining up of place values when multiplying or dividing vertically by omitting or forgetting to include zeros as placeholders in the partial products or quotients.
- Students may forget that common denominators are not necessary for multiplying or dividing fractions.
- Students may incorrectly assume that multiplication results in a product that is larger than the two factors. Instruction continues with students assessing the reasonableness of their answers by determining if the product will be greater or less than the factors within the given context.
- Students may incorrectly assume that division results in a quotient that is smaller than the dividend. Instruction continues with students assessing the reasonableness of their answers by determining if the quotient will be greater or less than the dividend within the given context.

Strategies to Support Tiered Instruction

- Instruction includes using visual models to illustrate and make meaning of situations represented in word problems.
- Instruction includes the use of estimation to ensure the proper placement of the decimal point in the final product or quotient of decimals.
 - For example, if finding the product of 12.3 and 4.8, students should estimate the product to be close to 60, by using 12 and 5 as friendly numbers, then apply the decimal to the actual product of 123 and 48, which is 5904. Based on the estimate, the decimal should be placed after 59 to produce 59.04.
- Teacher encourages students with a firm understanding of multiplying and dividing fractions to convert decimal values into equivalent fractional forms and then perform the desired operation and convert the solution back to decimal form.

- Teacher encourages students with a firm understanding of multiplying and dividing decimals to convert fractional values into equivalent decimal forms and then perform the desired operation and convert the solution back to fractional form.
- Teacher provides opportunities for students to comprehend the context or situation by engaging in questions such as:
 - What do you know from the problem?
 - Can you create a visual model to help you understand or see patterns in your problem?
- Teacher provides graph paper to use while applying an algorithm for multiplying or dividing to keep numbers lined up and help students focus on place value.
- Instruction includes providing opportunities to reinforce place values with the use of base ten blocks or hundredths grids.
- Instruction includes the co-creation of a graphic organizer utilizing the mnemonic device S.I.R. (Same, Inverse Operation, Reciprocal) for dividing fractions, which encourages the use of correct mathematical terminology, and including examples of applying the mnemonic device when dividing fractions, whole numbers, and mixed numbers.
- Teacher provides students with flashcards to practice and reinforce academic vocabulary.
- Instead of multiplying by the reciprocal to divide fractions, an alternative method could include rewriting the fractions with a common denominator, dividing the numerators, and then dividing the denominators.
 - For example, $\frac{5}{6} \div \frac{3}{2}$ is equivalent to $\frac{5}{6} \div \frac{9}{6}$ which is equivalent to $\frac{5/9}{1}$ which is equivalent to $\frac{5}{9}$.
- Instruction includes the use of a three-read strategy. Students read the problem three different times, each with a different purpose.
 - First, read the problem with the purpose of answering the question: What is the problem, context, or story about?
 - Second, read the problem with the purpose of answering the question: What are we trying to find out?
 - Third, read the problem with the purpose of answering the question: What information is important in the problem?
- Instruction provides opportunities to assess the reasonableness of answers by determining if the product will be greater or less than the factors within the given context.
- Instruction provides opportunities to assess the reasonableness of answers by determining if the quotient will be greater or less than the dividend within the given context.

Instructional Tasks

Instructional Task 1 (MTR.3.1, MTR.6.1)

Janie is at the gas station. She has \$53.25 and buys a sandwich that costs \$7.68 and a drink for \$0.97.

Part A. After she buys the sandwich and drink, how much money will Janie have left?

Part B. Janie wants to buy 10 gallons of gas with the remaining money. What is the highest price per gallon that she can afford? Use words or numbers to show your work.

Instructional Task 2 (MTR.3.1, MTR.7.1)

For the bake sale at school next week, Tamsynn plans to bake brownies.

Part A. If each batch makes 12 brownies, and uses $1\frac{1}{2}$ cups of flour, how many brownies could she make if she has 18 cups of flour.

Part B. If she plans to sell each brownie for \$1.25, how much money will she earn for the bake sale if she sells all of her brownies?

Part C. If her goal is to earn \$200, how many more brownies should she make?

Instructional Items*Instructional Item 1*

Candy comes in $3\frac{1}{2}$ pound bags. At a class party, the boys in the class ate $2\frac{1}{4}$ bags of candy and the girls in the class ate $1\frac{1}{3}$ bags. How many pounds of candy did the class eat?

Instructional Item 2

Tina's SUV holds 18.5 gallons of gasoline. If she has 4.625 gallons in her car when she stops to fill it up. How much money will she spend to fill up her car if the current price for gas is \$2.57 per gallon?

**The strategies, tasks and items included in the BIG-M are examples and should not be considered comprehensive.*